CHG 569
Environmental Pollution Engineering

Module IV

(Air Pollution Legislation)

Air Pollution Regulations (Legislations/Laws)

Air pollution laws not new

(Air pollution initially was recognized more as a nuisance than as a threat to human health)

In 1306. Edward I of England prohibited the burning of sea coal (soft coal) in craftsman's furnaces because of the foul-smelling fumes produced
Earlier control laws were for aesthetic reasons

As the years passed, air pollution got worse, and yet it was still not widely recognized as a threat to human health, not until the problem reached crisis proportions.

Study the following attachments

History of Air Pollution in

- UK
- USA
- Nigeria

See documents attached to lecture material

Laws exist at the Local

State

Federal, and

International levels (e.g. The Montreal Protocol of 1987 – to protect the ozone layer) and

Kyoto Protocol of 1997 to deal with ghgs

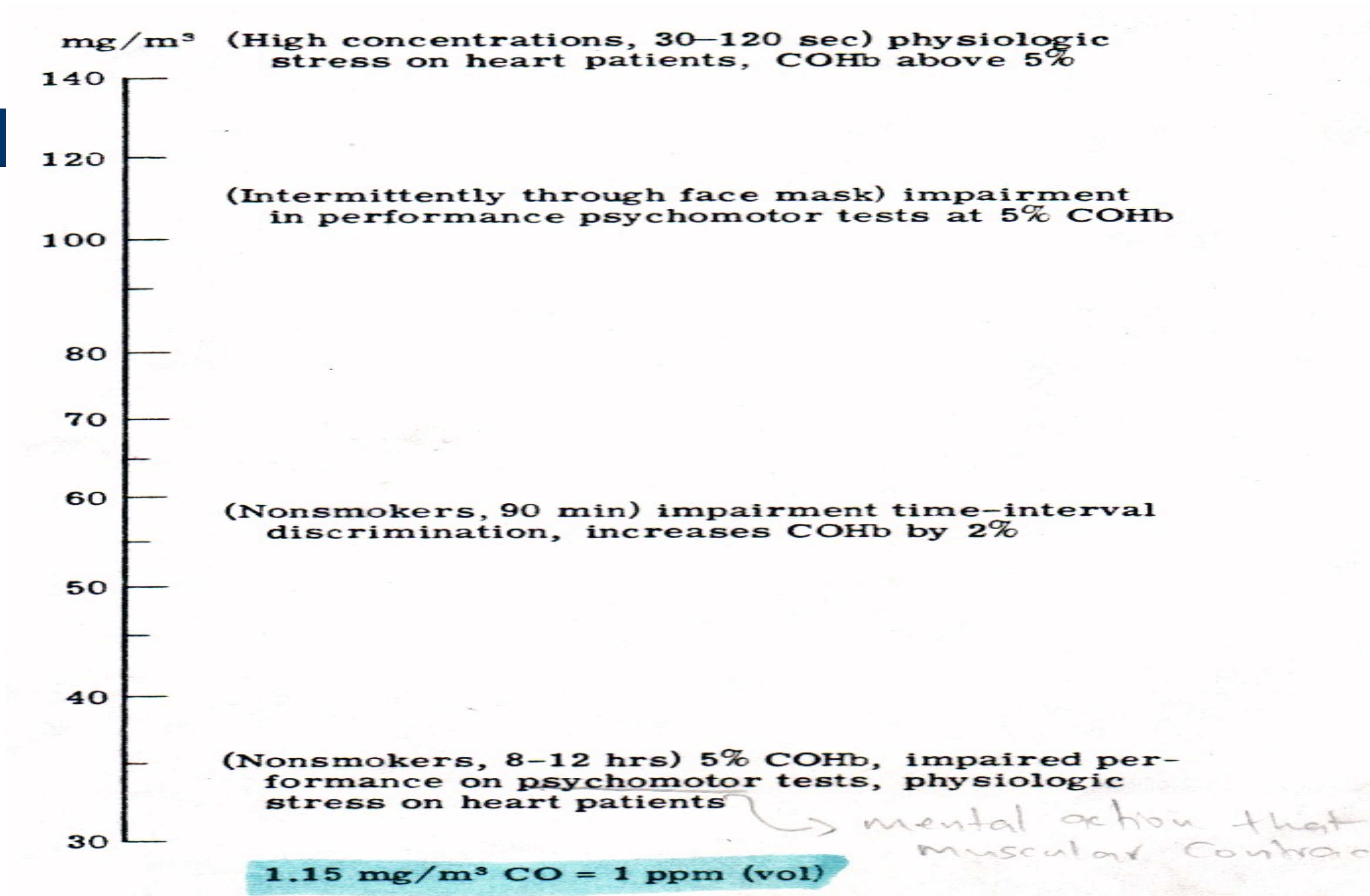
2 primary things legislation contains

- Emission standards
  - Air quality criteria (descriptive)- The levels of pollution and lengths of exposure above which adverse health and welfare effects may occur- must be set before standards can be determined
    - Primary – levels that protect human health
    - Secondary – levels that that will prevent other adverse effects

The six criteria (common) air pollutants (US-based) are:

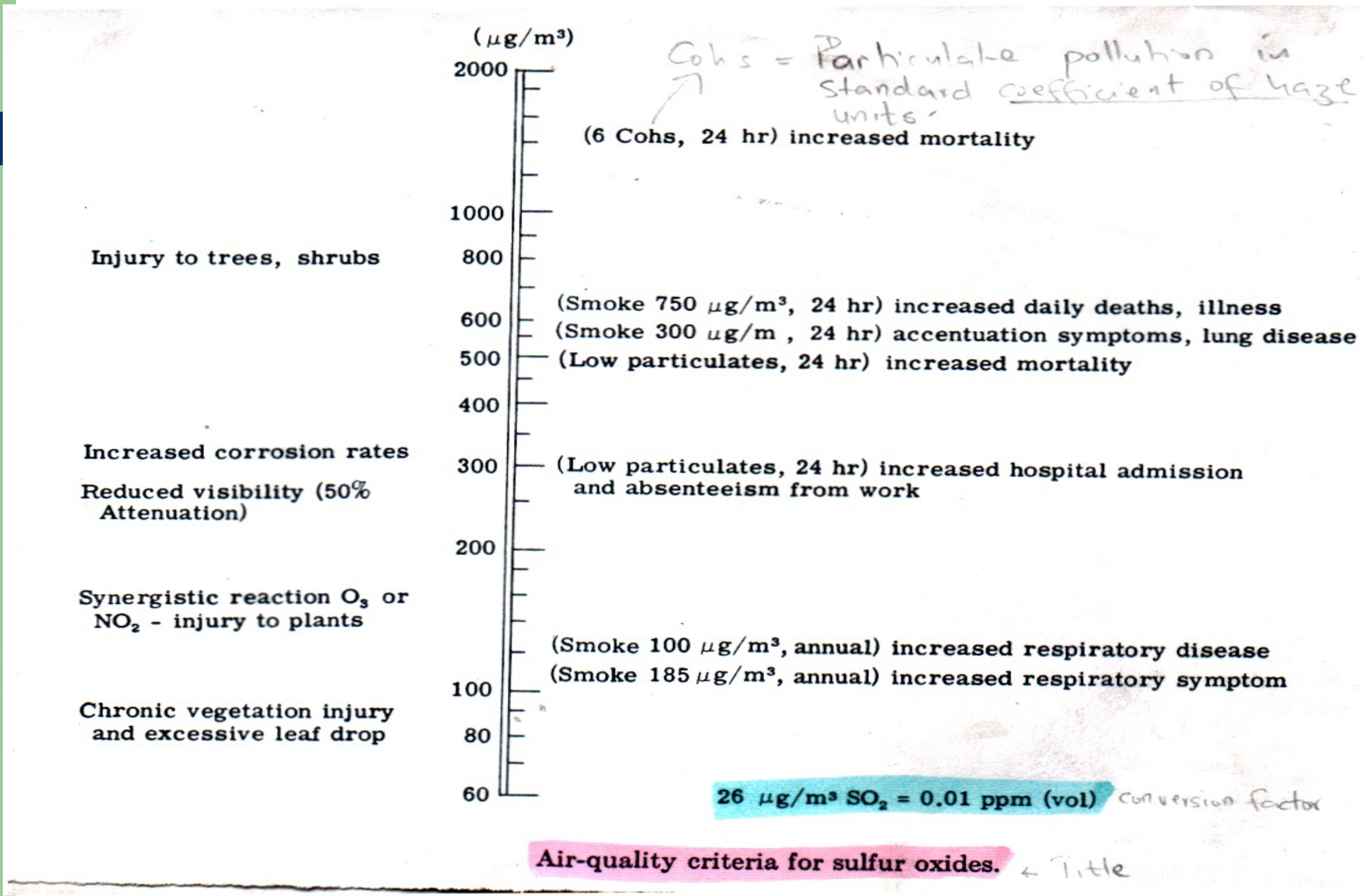
Carbon monoxide
Lead
Nitrogen oxides
Particulate Matter
Sulphur dioxide
Ozone (Photochemical smog)

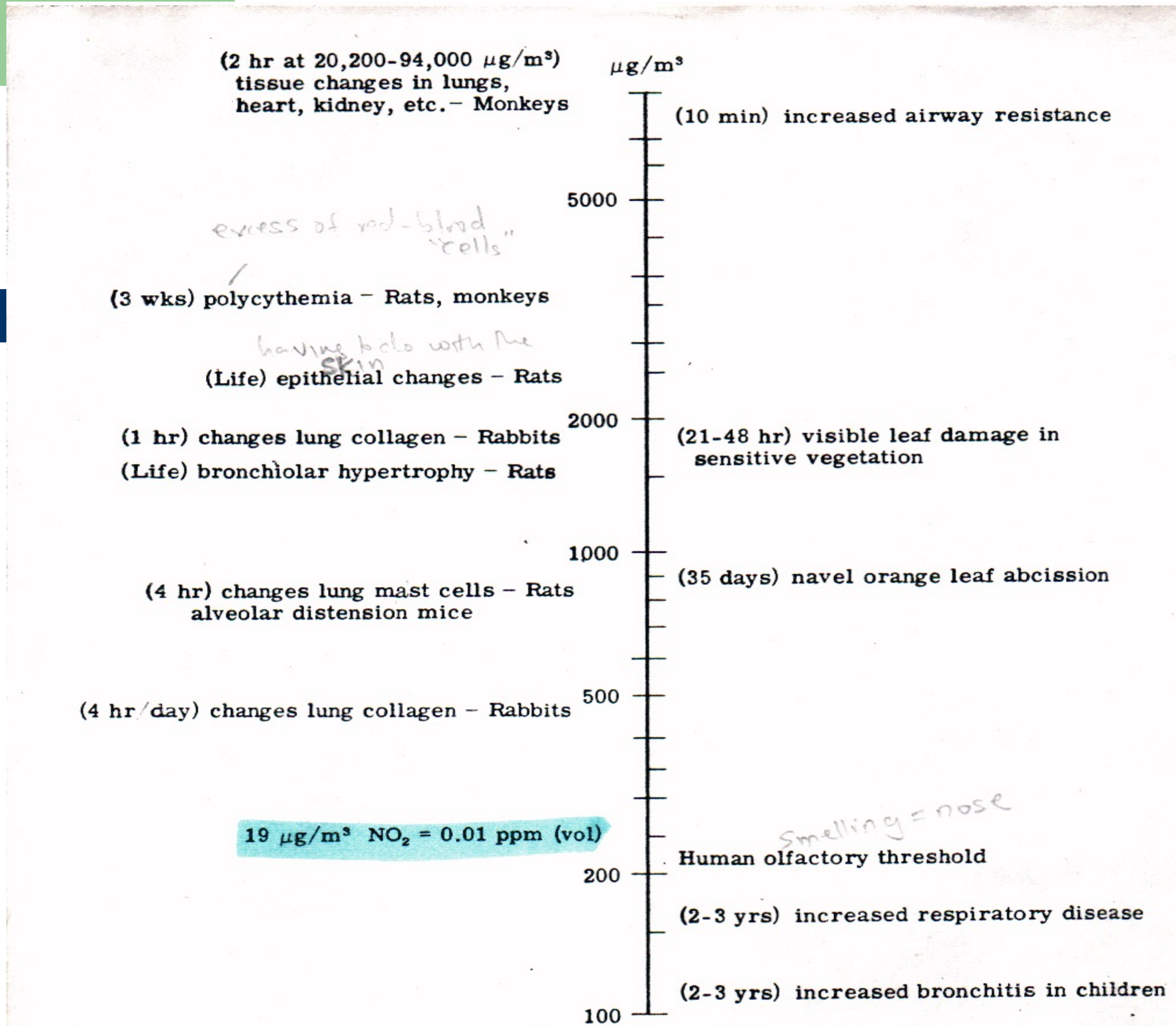
Air quality criteria for carbon monoxide



Air-quality criteria for carbon monoxide.

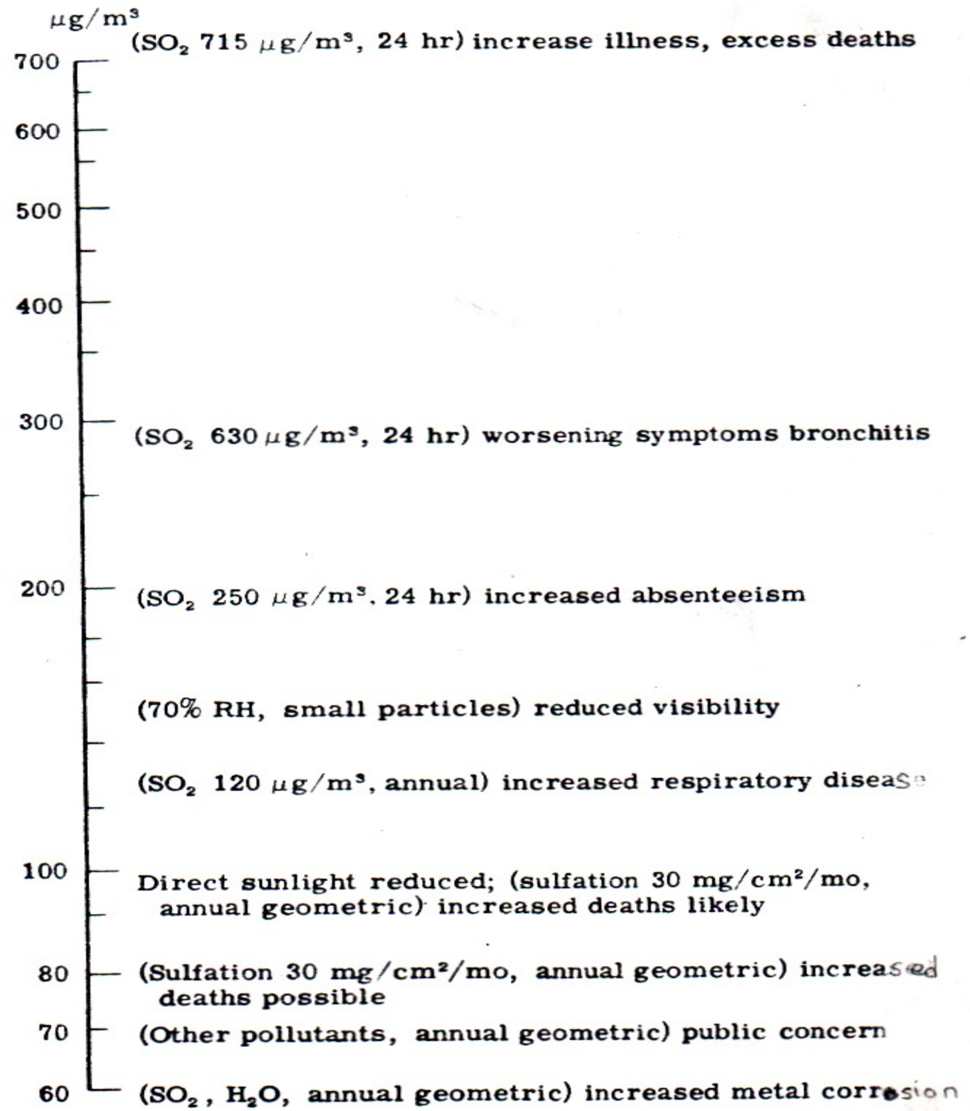
Air quality criteria for sulphur dioxide





Air-quality criteria for nitrogen dioxide.

Air quality criteria for Particulates



Air-quality criteria for particulates.

National Ambient Air Quality Standards, NAAQS

Pollutant [final rule cite]		Primary/ Secondary	Averaging Time	Level	Form
Carbon Monoxide [76 FR 54294, Aug 31, 2011]		primary	8-hour	9 ppm	Not to be exceeded more than once per year
			1-hour	35 ppm	
Lead [73 FR 66964, Nov 12, 2008]		primary and secondary	Rolling 3 month average	0.15 $\mu g/m^3$ [(1)]	Not to be exceeded
Nitrogen Dioxide [75 FR 6474, Feb 9, 2010] [61 FR 52852, Oct 8, 1996]		primary	1-hour	100 ppb	98th percentile, averaged over 3 years
		primary and secondary	Annual	53 ppb [(2)]	Annual Mean
Ozone [73 FR 16436, Mar 27, 2008]		primary and secondary	8-hour	0.075 ppm [(3)]	Annual fourth-highest daily maximum 8-hr concentration, averaged over 3 years
Particle Pollution Dec 14, 2012	$PM_{2.5}$	primary	Annual	12 $\mu g/m^3$	annual mean, averaged over 3 years
		secondary	Annual	15 $\mu g/m^3$	annual mean, averaged over 3 years
		primary and secondary	24-hour	35 $\mu g/m^3$	98th percentile, averaged over 3 years
	PM_{10}	primary and secondary	24-hour	150 $\mu g/m^3$	Not to be exceeded more than once per year on average over 3 years
Sulfur Dioxide [75 FR 35520, Jun 22, 2010] [38 FR 25678, Sept 14, 1973]		primary	1-hour	75 ppb [(4)]	99th percentile of 1-hour daily maximum concentrations, averaged over 3 years
		secondary	3-hour	0.5 ppm	Not to be exceeded more than once per year

as of October 2011

Recommended 2021 AQG levels compared to 2005 air quality guidelines

Pollutant	Averaging Time	2005 AQGs	2021 AQGs
$PM_{2.5}$, µg/m^3	Annual	10	5
	24-hour[a]	25	15
PM_{10}, µg/m^3	Annual	20	15
	24-hour[a]	50	45
O_3, µg/m^3	Peak season[b]	-	60
	8-hour[a]	100	100
NO_2, µg/m^3	Annual	40	10
	24-hour[a]	-	25
SO_2, µg/m^3	24-hour[a]	20	40
CO, mg/m^3	24-hour[a]	-	4

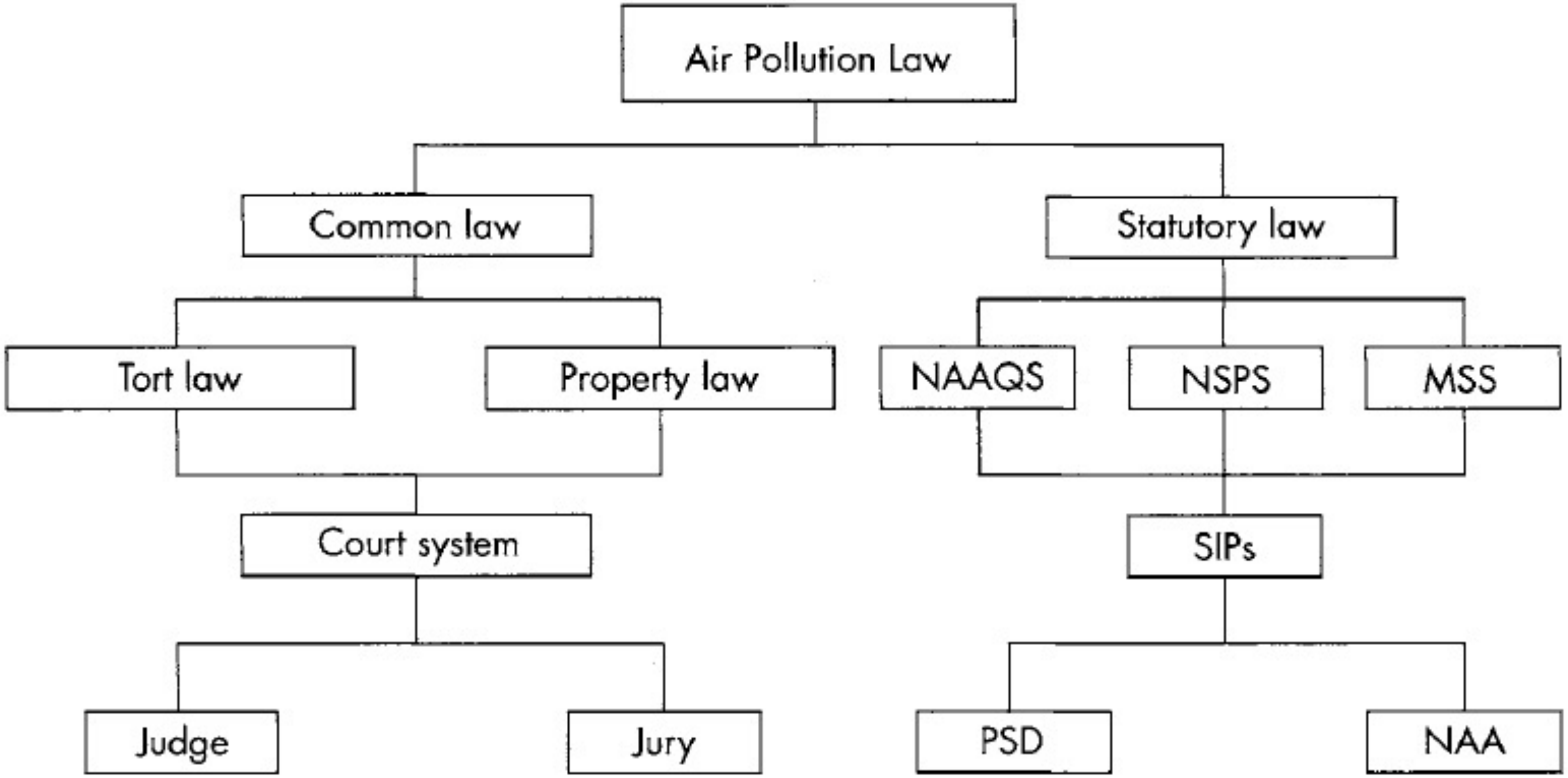
µg = microgram

[a] 99th percentile (i.e. 3–4 exceedance days per year).

[b] Average of daily maximum 8-hour mean O_3 concentration in the six consecutive months with the highest six-month running- average O_3 concentration.

Note: Annual and peak season is long-term exposure, while 24 hour and 8 hour is short-term exposure.

Air Pollution Law and Regulations in the US



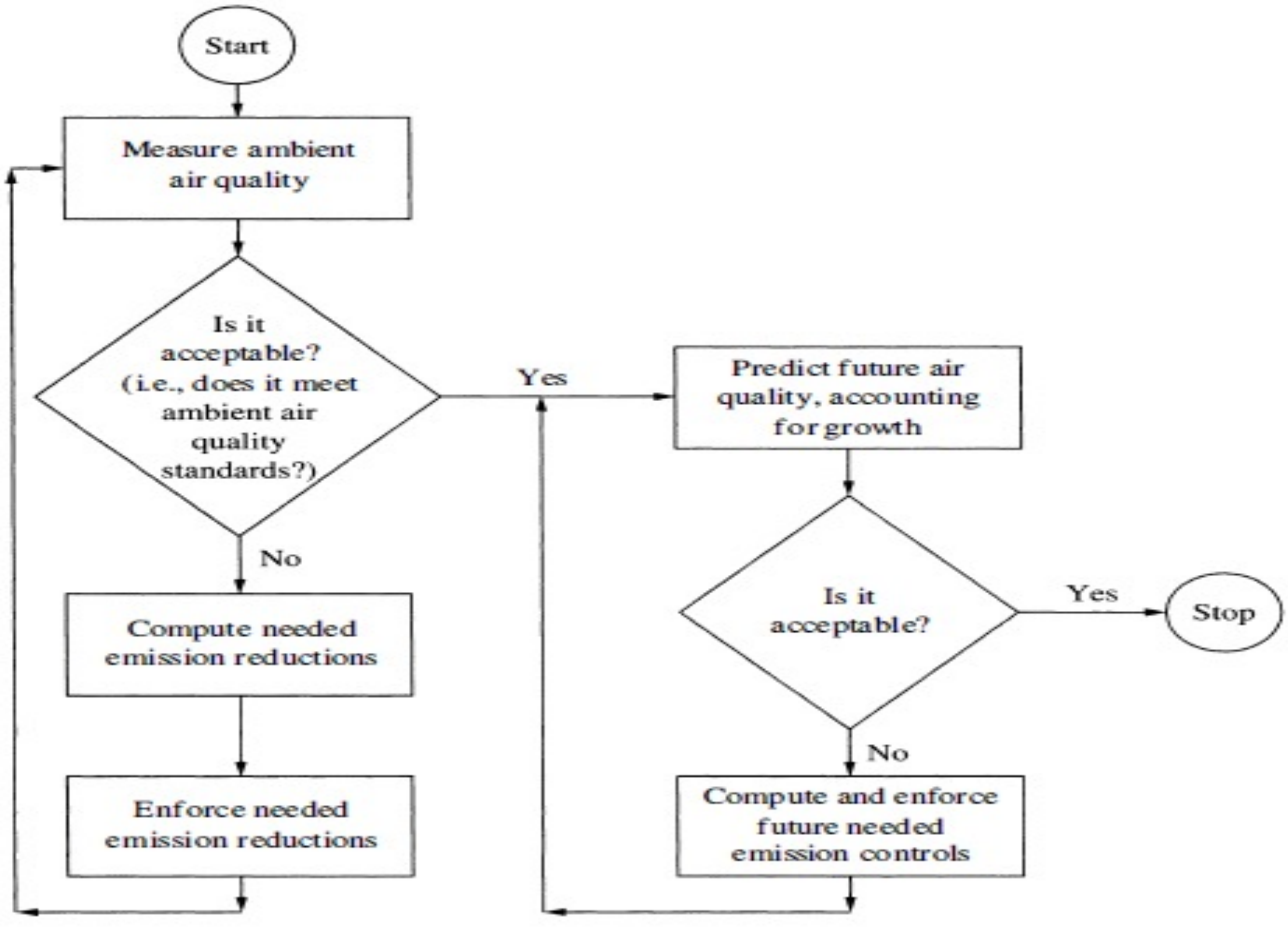
Key: NAAQS – National Ambient Air Quality Standards
NSPS – New (stationary) Source Performance Standards
MSS – Moving Source Standards
SIPs – State Implementation Plans
PSD – Prevention of Significant Deterioration
NAA – Nonattainment Areas

2 primary things legislation contains

- **Abatement strategy** framework to be used to control pollution – function of the nature of the pollutant.
  - **In the US**
    - **M**aximum **A**vailable **C**ontrol **T**echnology, MACT for HAPs (hazardous pollutants)
    - **B**est **A**vailable **C**ontrol **T**echnology, BACT for new plants
    - BAT - Best available technology
    - LAER - Lowest Achievable Emissions Rate,
    - RACT - Reasonably Available Control Technology
  - **In the UK**
    - BPM - best practicable means
    - BATNEEC - Best Available Technology Not Entailing Excessive Cost,

      ,
  - **In Nigeria**
    - BAPPT – Best Available and Practically Proven Technology

Protecting the environment thru legislation is a continuous process



The Nigerian Approach

to tackling air pollution

The **National Environmental Standards and Regulations Enforcement Agency (NESREA),** overseen by the Federal Ministry of Environment, is the main body responsible for enforcing environmental laws, standards, and regulations in Nigeria, including those related to air quality.

Nigeria has several regulations aimed at controlling air pollution, such as:

1. National Environmental (Air Quality Control) Regulations: This regulation provides the legal framework for the control of air pollution and the protection of air quality in Nigeria. It sets out standards and guidelines for emissions from various sources.
2. National Policy on the Environment: This policy provides a broad framework for environmental protection and conservation, including air quality management. It emphasizes sustainable development, pollution prevention, and the integration of environmental considerations into the developmental process.
3. National Guidelines and Standards for Environmental Pollution Control in Nigeria: These guidelines cover a wide range of environmental issues, including air pollution. They set out the acceptable limits of air pollutants and outline the responsibilities of governmental bodies, industries, and individuals in maintaining air quality.

However, despite these regulations, Nigeria faces significant challenges in enforcing environmental laws and reducing air pollution. Issues such as industrial emissions, vehicular emissions, open burning of waste, and the use of generators due to unreliable electricity supply are major sources of air pollution.